

| Question         |                   |                     |  | Marking details                                                                                                                                                                                         |                                                       |           | Marks available |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|------------------|-------------------|---------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------|-----------------|-------------------|---------------------|------------------|-------|------|-----------------|----|----|--|---|--|---|--|--|
|                  |                   |                     |  |                                                                                                                                                                                                         |                                                       |           | AO1             | AO2               | AO3                 | Total            | Maths | Prac |                 |    |    |  |   |  |   |  |  |
| 7                | (a)               |                     |  | salt <b>A</b>                                                                                                                                                                                           | potassium iodide                                      | accept KI | 2               |                   | 1                   | 3                |       | 3    |                 |    |    |  |   |  |   |  |  |
|                  |                   |                     |  | observation <b>B</b>                                                                                                                                                                                    | brick red (flame)                                     |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|                  |                   |                     |  | observations <b>C</b>                                                                                                                                                                                   | apple green (flame) <b>and</b><br>cream (precipitate) |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|                  |                   |                     |  | award (1) for each correct row                                                                                                                                                                          |                                                       |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|                  | (b)               |                     |  | <table><tr><td>ion</td><td>Number of protons</td><td>Number of electrons</td></tr><tr><td>Ca<sup>2+</sup></td><td>20</td><td>18</td></tr><tr><td>Br<sup>-</sup></td><td>35</td><td>36</td></tr></table> |                                                       |           | ion             | Number of protons | Number of electrons | Ca <sup>2+</sup> | 20    | 18   | Br <sup>-</sup> | 35 | 36 |  | 2 |  | 2 |  |  |
| ion              | Number of protons | Number of electrons |  |                                                                                                                                                                                                         |                                                       |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
| Ca <sup>2+</sup> | 20                | 18                  |  |                                                                                                                                                                                                         |                                                       |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
| Br <sup>-</sup>  | 35                | 36                  |  |                                                                                                                                                                                                         |                                                       |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|                  |                   |                     |  | award (1) for each correct row or column<br>no ecf from incorrect number of protons                                                                                                                     |                                                       |           |                 |                   |                     |                  |       |      |                 |    |    |  |   |  |   |  |  |
|                  | (c)               |                     |  | Ca <sup>2+</sup> + CO <sub>3</sub> <sup>2-</sup> → CaCO <sub>3</sub><br><br>award (1) for reactants<br>award (1) for product (as long as an equation has been attempted)<br><br>ignore state symbols    |                                                       |           | 2               |                   |                     | 2                |       |      |                 |    |    |  |   |  |   |  |  |